Step 1 – Sequence Retrieval and BLAST

The goal of this step is to investigate whether humans have a functional GULO gene and compare it across other species.

1. Use the rat GULO gene as a reference.
2. Confirm if humans produce a functional GULO protein.
3. Collect Sequences from all species in your data set for later analysis.

Part A: BLASTn (nucleotide comparison)

1. What to do?
   a. Take the rat GULO coding sequence (mRNA).
   b. Run BLASTn against the human genome (GRCh38).
2. Why?
   a. Look for mismatches, gaps, stop codons that might indicate the human gene is non-functional.
3. Expected results?
   a. In BLAST
      Query: Rat GULO coding seq
      Database: Human Genome

      BLAST will locate sequence in human genome that looks like rat GULO (hits).
      Quantifies similarities (E-value, Alignment coverage…)
      You get a table: %identity, mismatches, gaps, e-value…
   b. The human locus on chromosome 8 aligns partially but will likely show mutations, INDELS, or premature stop codons that indicate it's a pseudogene.
4. Step by step
   1) Go to the NCBI Gene database: https://www.ncbi.nlm.nih.gov/gene
   2) In the search bar, type:
      GULO Rattus norvegicus
   3) Select the gene entry for **gulonolactone oxidase (GULO)** in rat.
   4) Scroll to the **NCBI Reference Sequences (RefSeq)** section and click on the mRNA accession (e.g., NM_ entry).
   5) Click on **FASTA** to display the sequence.
   6) Copy the **entire FASTA sequence**, including the header line beginning with >.
   7) Go to the NCBI BLAST page: https://blast.ncbi.nlm.nih.gov/Blast.cgi
   8) Select **Nucleotide BLAST (BLASTn)**.
   9) Paste the copied rat GULO FASTA sequence into the **"Enter Query Sequence"** box.
   10) Under **Database**, select:
      Human genomic + transcript (GRCh38)

11) In the **Organism** field, type and select:
    Homo sapiens
12) Leave all other parameters as default (Megablast).
13) Click **BLAST** to run the analysis.
14) Once results are generated, go to the **Descriptions** tab and record:
    - Query cover
    - Percent identity
    - E-value
    - Top hit description

5. Results and interpretation

RatGULO_Human_BLASTn_Description.cvs

Column	Value	Interpretation
Description	Homo sapiens chromosome 8, GRCh38.p14 Primary Assembly	The sequence in the human genome that matched the rat GULO query is located on **chromosome 8**, specifically in the primary assembly version **GRCh38.p14**. This indicates the putative **human GULO locus**.
Scientific Name	Homo sapiens	Confirms that the subject sequence is from humans.
Max Score	119	The highest alignment score for a single local region between the rat GULO query and the human sequence. **Higher scores indicate better similarity**; a score of 119 is modest for a coding gene (~1–1.5 kb), suggesting **limited similarity**.
Total Score	119	Sum of all alignment scores for this subject. Since it equals the Max Score, **there is only one alignment region**, indicating the human locus only partially matches the rat GULO gene.
Query Cover	7%	Only **7% of the rat GULO coding sequence aligns** to the human genome. This is extremely low coverage, suggesting the **human GULO sequence is highly degraded or fragmented**, consistent with a pseudogene.
E-value	6e-24	The **expected number of matches by chance** is extremely low, confirming that this alignment is statistically significant and not random.

Per. ident	88.66%	The aligned region is **~89% identical** at the nucleotide level. This is relatively high for a small region, indicating the sequence is conserved in this fragment, but overall low coverage (7%) means the gene is mostly missing or heavily mutated in humans.
Acc. Len	145,138,636	The length of the human chromosome 8 sequence is 145 Mbp. The aligned region is a **tiny fraction** of this chromosome.
Accession	NC_000008.11	Reference accession for the human chromosome 8 sequence in **GenBank**. Useful for retrieving the exact genomic region for further analysis.

Low Query Coverage (7%)

- Only a small fragment of the rat GULO gene aligns with the human genome.
- Indicates that the human GULO gene is **largely absent or disrupted**, consistent with it being a **pseudogene**.

Moderate Percent Identity (88.66%) in the aligned fragment

- The small aligned region is fairly conserved, suggesting **remnant sequence conservation**, likely because it was functional in a common ancestor.
- However, the small coverage means most of the gene has **mutations, deletions, or insertions** that prevent functional expression.

Max/Total Score

- Only one significant alignment region is detected.
- Reinforces that the gene is **fragmented** and there are no additional intact regions in humans that resemble rat GULO.

E-value (6e-24)

- Extremely low, meaning the alignment is **highly unlikely to occur by chance**, even though it covers only 7% of the gene.
- Confirms that the aligned fragment is **homologous** to rat GULO.

Chromosomal Location (Chr 8)

- Matches previous studies: the human GULO pseudogene (GULOP) is located on **chromosome 8**, and this aligns perfectly with the reference genome.

RatGULO_Human_BLASTn_Alignments.txt

```
Query  229       CAGGTGGACAAGGAGAAGAAGCAGGTAACAGTGGAAGCCGGTATCCTCCTGGCTGACCTG  288
                 ||||||||  |||||||||||||||| ||  |||| || || ||||||||||||  |||||
Sbjct  27560277  CAGGTGGATGAGGAGAAGAAGCAGGTGACCATGGAGGCTGGCATCCTCCTGGCCAACCTG  27560336

Query  289       CACCCACAGCTGGATGAGCATGGCCTGGCCATGTCCA  325
                 ||||||||||||||||||||||||||||||| ||||||
Sbjct  27560337  CACCCACAGCTGGATGAGCATGGCCTGGCCCTGTCCA  27560373
```

Mismatch

- In the rat, the A mismatching the G is likely critical for the protein to fold correctly. In humans, because the gene is "dead," a mutation changed it to a G and nobody "fixed" it because we don't use the gene anyway.
- These single nucleotide changes can alter codons, potentially introduce amino acid changes or stop codons.
- This region is partially conserved, but differences suggest loss of perfect functionality if translated.

Functional implication

- These short, partially conserved fragments indicate that the human GULO locus is non-functional.
- This alignment is likely part of the human GULOP pseudogene, not an active coding sequence.

Evolutionary insight

- Conservation in these fragments suggests that the gene was functional in a common ancestor, but mutations accumulated in humans, resulting in gene loss of function.
- If we didn't share an ancestor with rats, we wouldn't find this specific sequence on our Chromosome 8 at all. The fact that it's there—but broken—proves we "lost" the ability rather than never having it.

Part B: BLASTp (Protein comparison of GULO)

1. What to do?
    a. Retrieve the rat GULO protein sequence (functional enzyme).
    b. Run BLASTp using this sequence as a query against the human proteome.
    c. Analyze whether any human protein significantly matches GULO.
2. Why?
    a. This answers the question: “Even if we have a gene-like sequence in DNA, does it produce a functional protein?”
    b. This also helps us bypass DNA ambiguity where only translated, functional genes produce proteins.
    c. To confirm loss of function. If not protein match exists, this means that the gene is not expressed properly, or the gene is non-functional.
3. Step by step

1) Go to: https://www.ncbi.nlm.nih.gov/protein
2) Search:
   GULO Rattus norvegicus
3) Select the correct gulonolactone oxidase protein entry
4) Click FASTA
5) Copy the full sequence (including header)
6) Go to: https://blast.ncbi.nlm.nih.gov/Blast.cgi
7) Click **Protein BLAST (BLASTp)**
8) Paste the rat GULO protein FASTA sequence into:
   "Enter Query Sequence"
9) Under Database, choose:
   non-redundant protein sequences (nr)
10) Under Organism, type and select:
    Homo sapiens
11) Click BLAST and wait for results

4. Results and interpretation

RatGULO_Human_BLASTp_Descriptions.csv

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. ident
24-dehydrocholesterol reductase [Homo sapiens]	Homo sapiens	65.1	65.1	27%	6e-10	29.41
delta(24)-sterol reductase precursor [Homo sapiens]	Homo sapiens	65.1	65.1	27%	6e-10	29.41
putative protein product of Nbla03646 [Homo sapiens]	Homo sapiens	64.3	64.3	27%	7e-10	29.41
unnamed protein product [Homo sapiens]	Homo sapiens	63.9	63.9	27%	1e-09	29.41
unnamed protein product [Homo sapiens]	Homo sapiens	63.9	63.9	27%	1e-09	29.41
24-dehydrocholesterol reductase, partial [Homo sapiens]	Homo sapiens	50.4	50.4	22%	2e-06	30.93
alkyldihydroxyacetonephosphate synthase, peroxisomal isoform X1 [Homo sapiens]	Homo sapiens	46.2	46.2	40%	5e-04	26.15
alkyldihydroxyacetonephosphate synthase, peroxisomal precursor [Homo sapiens]	Homo sapiens	45.8	45.8	40%	8e-04	26.15
alkylglycerone phosphate synthase, isoform CRA_a [Homo sapiens]	Homo sapiens	45.4	45.4	40%	0.001	26.15

alkylglycerone phosphate synthase, isoform CRA_b [Homo sapiens]	Homo sapiens	45.4	45.4	40%	0.001	26.15
unnamed protein product [Homo sapiens]	Homo sapiens	44.7	44.7	40%	0.001	26.63
unknown, partial [Homo sapiens]	Homo sapiens	39.7	39.7	19%	0.048	31.87

Sequence Similarity Analysis

- The percent identity of the alignments ranges from **~26% to 31%**.
- These values indicate very low sequence similarity.
- In protein comparisons:
  - 40–50% identity suggests strong functional similarity
  - 30–40% suggests weak similarity
  - <30% suggests minimal or random similarity
- The observed values (~29%) fall within the low similarity range, indicating no meaningful homology.

Query Coverage

- Query coverage ranges from **22% to 40%**.
- This indicates that only a small portion of the rat GULO protein aligns with human proteins.
- The low coverage suggests the absence of a full-length homologous protein in humans.

E-value Interpretation

- E-values range from **6×10^{-10} to 0.048**.
- Lower E-values indicate statistically significant matches.
- However, in this context:
  - The low E-values correspond to short conserved regions, not full protein similarity.
- These matches likely reflect shared enzymatic motifs, not functional equivalence.

Alignment Scores

- Max scores range from approximately **40 to 65**.
- These are low alignment scores for protein BLAST.
- Functional homologs typically produce much higher scores.
- This further supports the weak nature of the matches.

Functional Interpretation of Hits

- The proteins identified (e.g., sterol reductases, peroxisomal enzymes) are:
  - **Oxidoreductases**, similar to GULO in general function

    - Likely to share **common structural or catalytic domains**
- The observed similarity is therefore generic enzymatic similarity not specific to GULO function.

Absence of GULO in Humans

- No BLASTp hit corresponds to a human GULO protein.
- This indicates that:
    - The human genome does not encode a functional GULO enzyme
    - Any GULO-like sequence present at the DNA level is not translated into a functional protein

Part C: Sequence Retrieval of all Species

1. What to do?
    a. Retrieve the GULO sequences for all species in the dataset.
    b. For each species collect: coding DNA sequence and protein sequence.
    c. Store all sequences in FASTA format.
2. Why?
    a. This prepares our data for future applications in the project (comparative analysis, phylogenetic trees…).